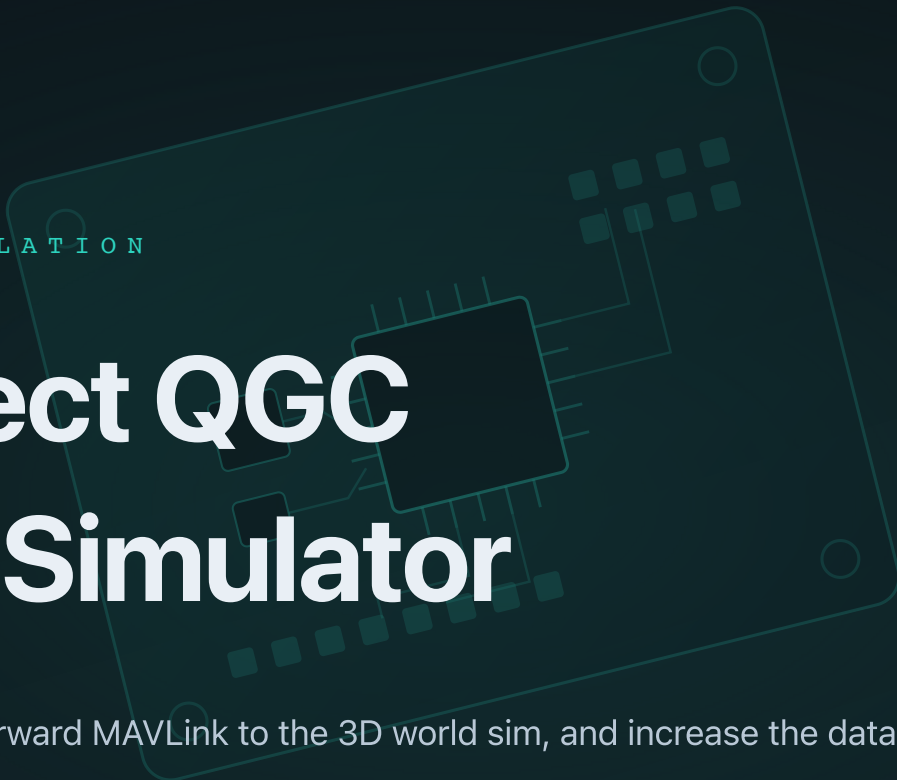




DAY 5 • SIMULATION

Connect QGC to the Simulator

Link QGC to SITL, forward MAVLink to the 3D world sim, and increase the data rate.

~10 minutes

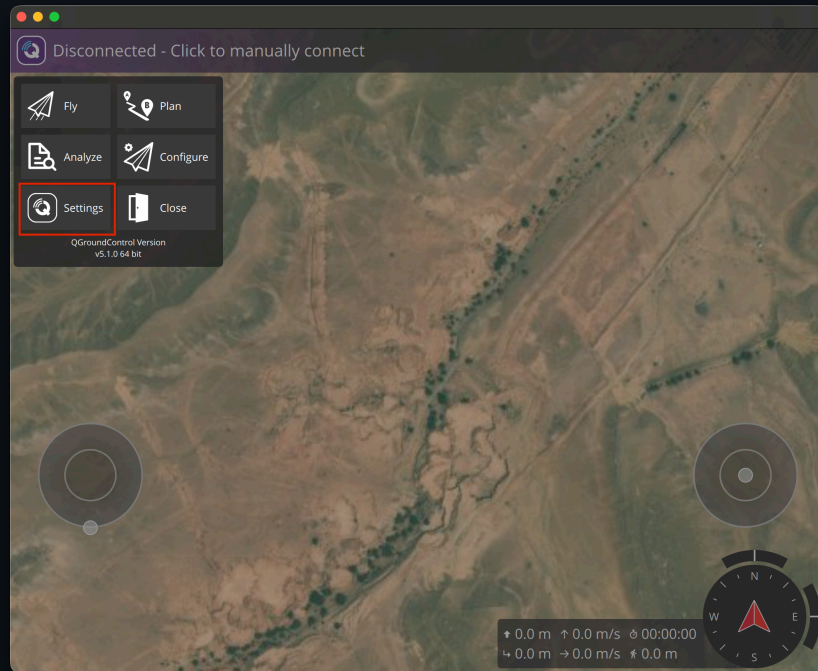
3 D S I M U L A T I O N

Connect QGC to the simulator

SITL runs the firmware. QGC gives you the GCS. A 3D world sim gives you eyes. Connect all three and you can fly, plan, and watch — without hardware.

Step 1 — Open Settings

Open QGC. Click the **Q** icon → [Settings](#).

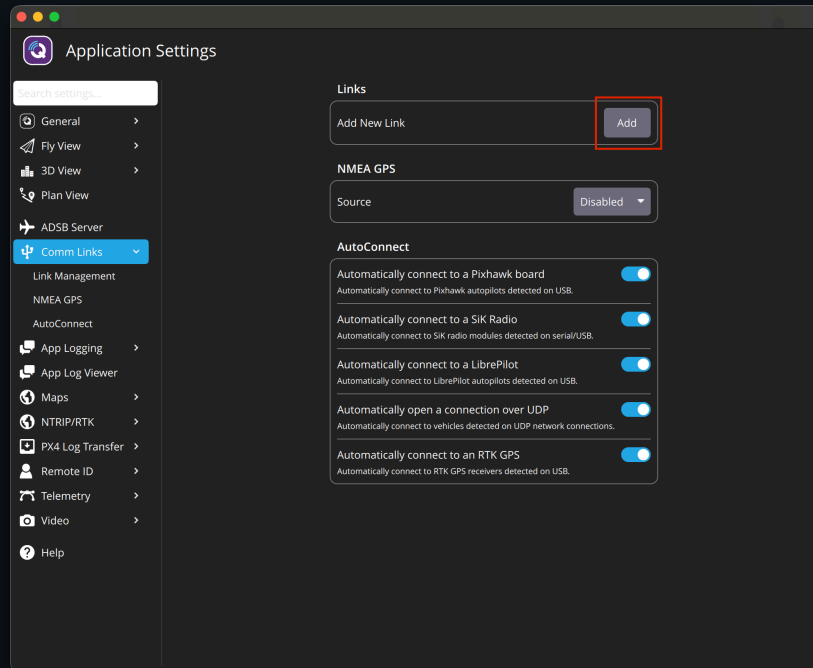


WHY SETTINGS FIRST

QGC needs to know **where the simulator is**. By default it auto-connects to USB and UDP — for SITL we add a manual TCP link.

Step 2 — Add a new link

Go to **Comm Links** in the sidebar. Click **Add**.

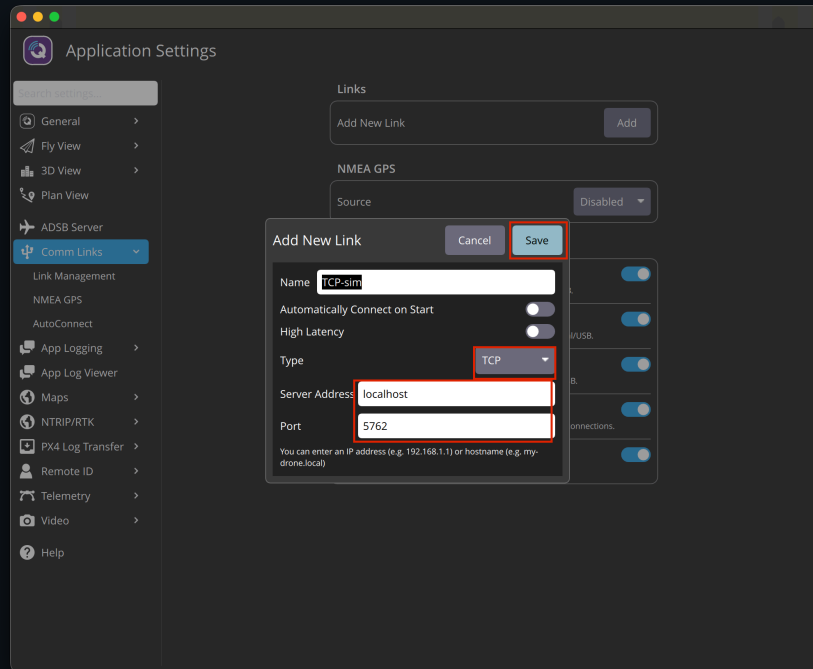


COMM LINKS

This page lists all the ways QGC can connect to a vehicle — USB, serial, UDP, TCP. Click **Add** to create a new TCP connection to the simulator.

Step 3 — TCP to localhost:5762

Fill in the connection details and click **Save**.



CONNECTION SETTINGS

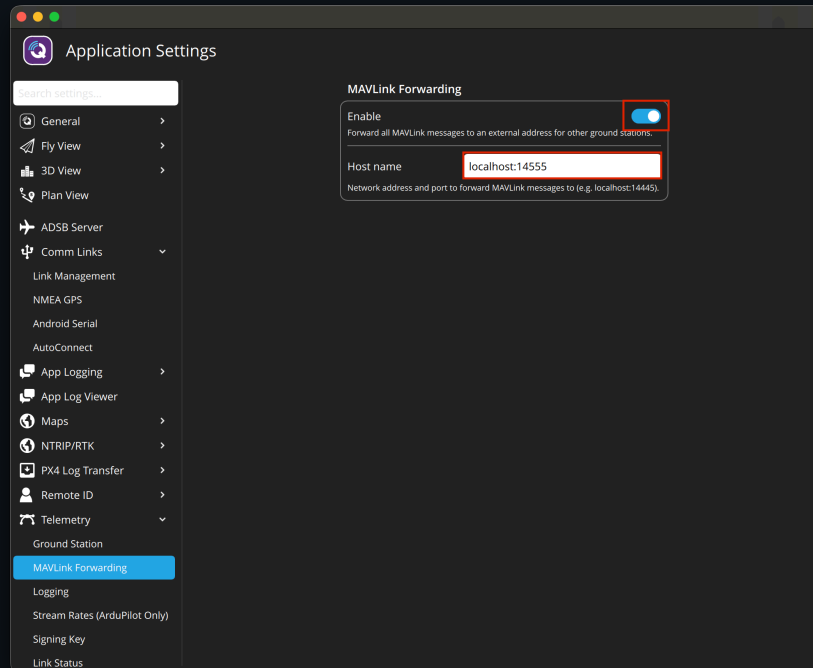
Type	TCP
Server Address	localhost
Port	5762

SAVE AND CONNECT

After saving, select the link and click **Connect**. QGC connects to SITL — you should see the HUD go live with Stabilize mode and 100% battery.

Step 4 — MAVLink forwarding to the world sim

Still in Settings → **MAVLink Forwarding**. Enable it and set the host.



FORWARD MAVLINK

Enable

On

Host name

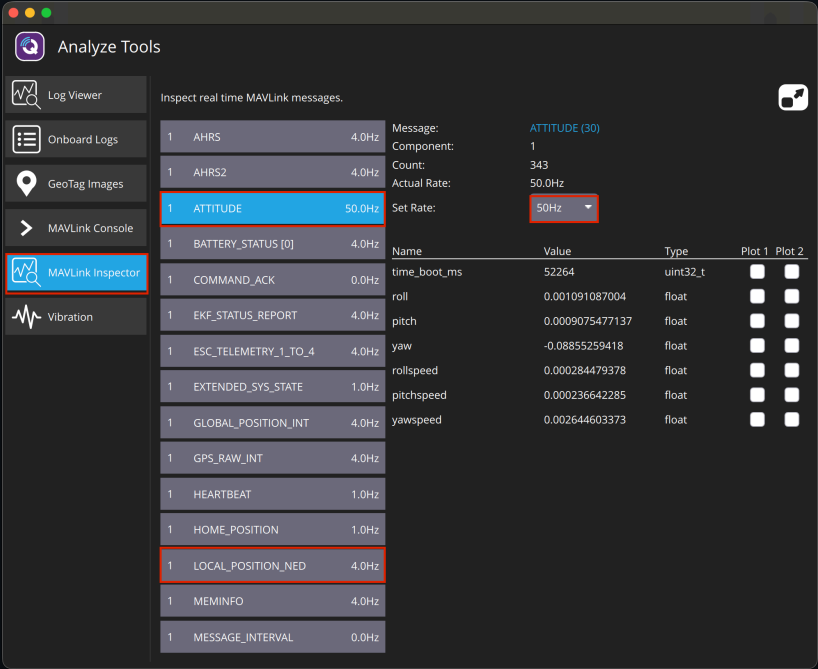
localhost:14555

WHAT THIS DOES

QGC forwards every MAVLink message it receives to `localhost:14555` — where the 3D world simulator is listening. The sim reads attitude and position messages to render the aircraft in 3D.

Step 5 — MAVLink Inspector — increase data rate

Click Q icon → Analyze → MAVLink Inspector.



INCREASE THE RATE

Find **ATTITUDE** and **LOCAL_POSITION_NED** in the message list. Set their rate to **50 Hz**. The default 4 Hz is too slow for smooth 3D rendering.

WHY THESE TWO

ATTITUDE (#30) — roll, pitch, yaw. The sim needs this to tilt the aircraft model. **LOCAL_POSITION_NED** (#32) — x, y, z position. The sim needs this to move it in 3D space.

THIS IS LIVE MAVLINK

The inspector shows you every message the aircraft is sending, in real time. This is the protocol from the MAVLink module — you are looking at it raw.

Attitude at 50 Hz, position at 50 Hz. Now start the world sim — it has everything it needs to render the aircraft in 3D.